Unit: 6 Unit 5: Linear Equations **Name:** ____________
Topic: Point-Slope Form, Slope-Intercept Form, Standard Form and Converting Forms **Date:** ________

Challenge (Habanero)

Q1. A recipe for making cookies calls for $2x$ cups of flour and $3x+5$ cups of sugar. If x represents the number of batches, which equation represents the total amount of ingredients in standard form?
(A) $5x+5=0$
(B) $5x-5=0$
(C) $2x+3x+5=0$
(D) $5x+5=y$
(E) $x-5=0$

Q2. The width of a rectangular garden is x meters and its length is $3x+2$ meters. If the perimeter is 20 meters, which equation represents this situation in slope-intercept form?
(A) $y=3x+2$
(B) $y=2x+3$
(C) $y=x+6$
(D) $x+y=10$
(E) $2x+2y=20$

Q3. A construction company is building a new bridge with an arch that rises 2 meters for every 5 meters of horizontal distance. If the equation of the arch is written in point-slope form, with a point at $(0,0)$, which of the following is the correct equation?
(A) $y-0=\frac{2}{5}(x-0)$
(B) $y-0=\frac{5}{2}(x-0)$
(C) $y=\frac{2}{5}x$
(D) $y=\frac{5}{2}x$
(E) $x-y=0$

Q4. A chef has a recipe for trail mix that includes $\frac{1}{4}$ cup of nuts for every $\frac{1}{2}$ cup of dried fruit. Which equation in slope-intercept form represents the amount of nuts (y) based on the amount of dried fruit (x)?
(A) $y=\frac{1}{2}x$
(B) $y=\frac{1}{4}x$
(C) $y=\frac{1}{2}x+1$
(D) $y=\frac{1}{4}x+\frac{1}{4}$
(E) $y=\frac{2}{1}x$

Q5. A building's height (y) in meters is related to the number of floors (x) by the equation $y=4x-2$. What is the y-intercept and what does it represent in this context?
(A) $(0,-2)$, the height of the building before construction starts
(B) $(0,2)$, the number of floors below ground level
(C) $(0,0)$, the ground level of the building
(D) $(-2,0)$, the number of floors before construction starts
(E) $(2,0)$, the height of the building above ground level

Q6. The cost (y) of producing x units of a product is given by the equation $y=5x+2000$. What is the slope of this equation and what does it represent?
(A) 5, the cost per unit of the product
(B) 2000, the initial cost of production
(C) 5, the number of units produced per day
(D) 2000, the cost of producing one unit
(E) 5, the total cost of production per day

Q7. A water tank is being filled at a rate of 2 cubic meters per minute. If x is the time in minutes and y is the amount of water in the tank, which equation in standard form represents the amount of water?
(A) $2x-y=0$
(B) $y-2x=0$
(C) $2x+y=0$
(D) $x+2y=0$
(E) $y=2x$

Q8. The equation of a line in point-slope form is $y-3=2(x-4)$. Which of the following is the equation in slope-intercept form?
(A) $y=2x-5$
(B) $y=2x+3$
(C) $y=2x-8+3$
(D) $y=2x+4$
(E) $y=x+3$

Open Ended Questions (FRQ)

Q9. A bakery is making a special cake that requires $2x+1$ cups of flour and $x-2$ cups of sugar, where x is the number of cakes being made. Write an equation that represents the total amount of ingredients needed and graph the equation. What is the slope and y-intercept of the equation and what do they represent in this context? Show all work and explain your answers. Provide the equation in slope-intercept form.

Q10. A contractor is building a new house with a roof that rises 3 meters for every 4 meters of horizontal distance. If the equation of the roof is written in standard form, with a point at $(0,0)$, what is the equation of the line? Graph the equation and find the x- and y-intercepts. What do the intercepts represent in this context? Provide the equation in standard form and show all work.

Chef's Tips

- For questions 1-8, choose the correct answer from the given options.
- For questions 9-10, show all work and explain your answers.
- Chef's Tips